

business sharks Lukas and Julien

Lukas and Julien

2025-11-03

```
## le package 'lubridate' a été décompressé et les sommes MD5 ont été vérifiées avec succès
##
## Les packages binaires téléchargés sont dans
## C:\Users\bruno\AppData\Local\Temp\RtmpsHC3uE\downloaded_packages

## le package 'readr' a été décompressé et les sommes MD5 ont été vérifiées avec succès
##
## Les packages binaires téléchargés sont dans
## C:\Users\bruno\AppData\Local\Temp\RtmpsHC3uE\downloaded_packages

## le package 'MASS' a été décompressé et les sommes MD5 ont été vérifiées avec succès
##
## Les packages binaires téléchargés sont dans
## C:\Users\bruno\AppData\Local\Temp\RtmpsHC3uE\downloaded_packages

## le package 'maps' a été décompressé et les sommes MD5 ont été vérifiées avec succès
##
## Les packages binaires téléchargés sont dans
## C:\Users\bruno\AppData\Local\Temp\RtmpsHC3uE\downloaded_packages
```

#Introduction

This study focuses on the excess returns generated by American politicians' stock trades. The main objective is to analyze the relationship between excess returns and net gains across political party, state, and individual politicians.

1) Organization of the data

```
## [1] "Ticker"           "TickerType"       "Company"
## [4] "Traded"           "Transaction"      "Trade_Size_USD"
## [7] "Status"           "Subholding"       "Description"
## [10] "Name"             "Filed"            "Party"
## [13] "District"         "Chamber"          "Comments"
## [16] "Quiver_Upload_Time" "excess_return"    "State"
## [19] "last_modified"
```

The dataset consists of 17 columns, as presented above. However, the raw data required reorganization and cleaning before analysis.

After preprocessing, several unnecessary columns were removed and missing column names were eliminated. Date variables were standardized across all observations. Trade size values reported as ranges were converted into minimum and maximum values, and a midpoint measure was constructed.

Additional variables were created, including:

Trade delay, measured in days between the trade date and filing date

Net trade direction, where purchases were coded as +1 and sales as -1

Midpoint trade size, representing the average of the reported range

Net gain, calculated as excess return multiplied by trade direction

```
## [1] "Ticker"           "Company"           "Traded"
## [4] "Transaction"      "Trade_Size_USD"    "Subholding"
## [7] "Description"      "Name"              "Filed"
## [10] "Party"            "District"          "Chamber"
## [13] "Comments"         "excess_return"     "State"
## [16] "last_modified"    "Trade_Size_USD_min" "Trade_Size_USD_max"
## [19] "trade_delay"      "net_trade_direction" "trade_size_mid"
## [22] "net_gain"
```

After reorganization, the dataset now contains 21 columns. Three original variables were removed and seven new variables were added. The most important constructed variables are net gain, midpoint trade size, and trade delay. Net gain reflects the economic outcome of each transaction by combining excess return with trade direction.

2) Trade Size and number

We examine both the size and the number of trades on a yearly and monthly basis and compute correlations between these measures.

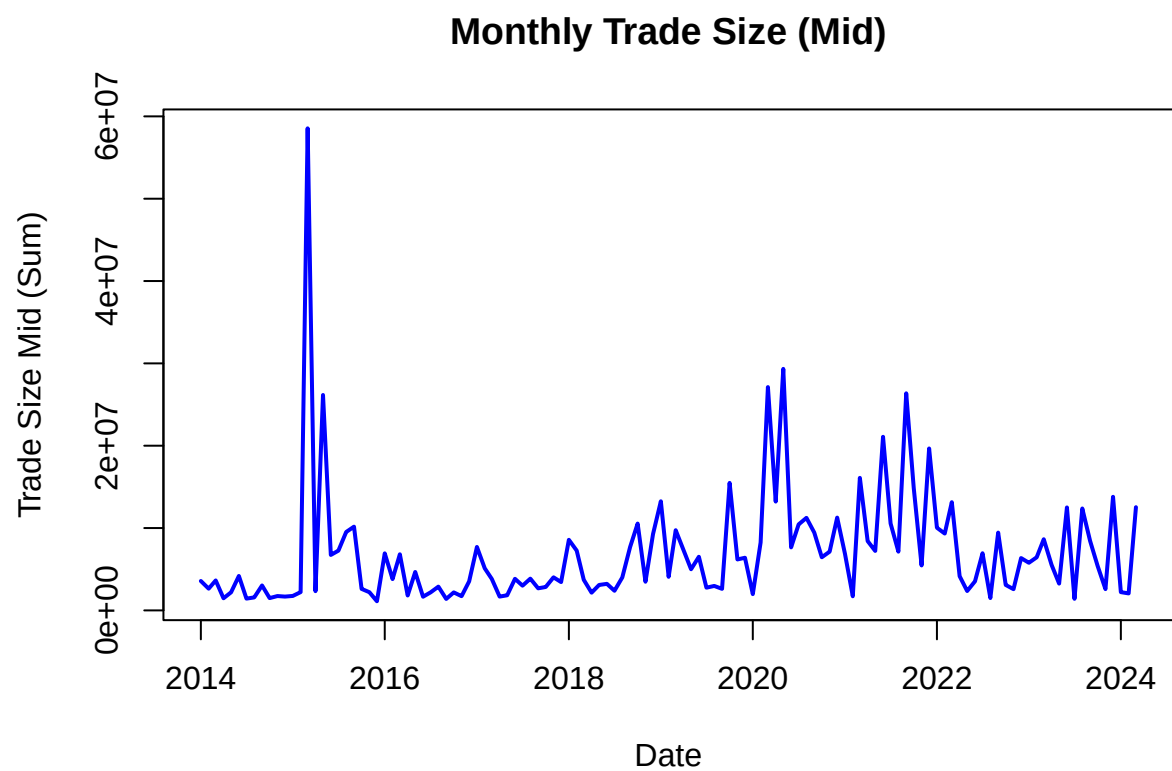
The mid size of trades per year is presented as follows:

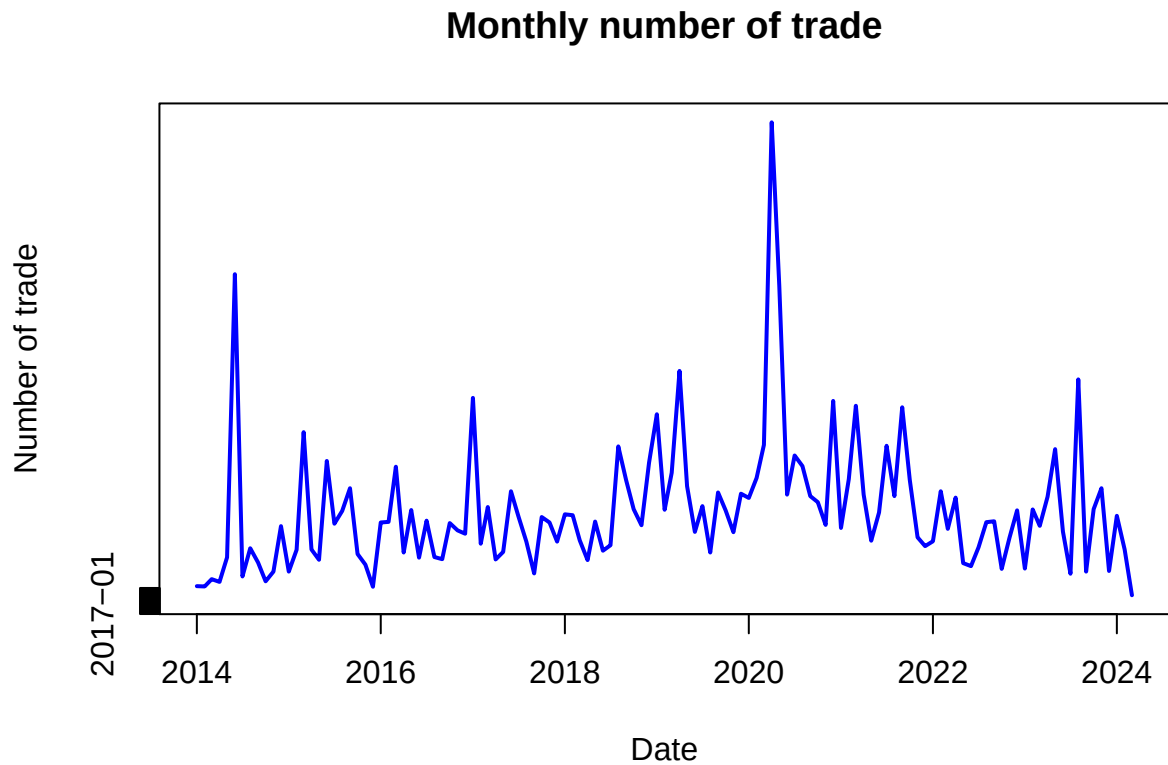
```
##      2014      2015      2016      2017      2018      2019      2020      2021
## 66154103 228706399 87088054 98730360 150494902 180750399 349174081 328230682
##      2022      2023      2024
## 170013831 206920068 45168016
```

While the number of trades per year is:

```
## years
## 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
## 3205 3797 3786 4087 4394 5377 7699 5377 3652 4383 710
```

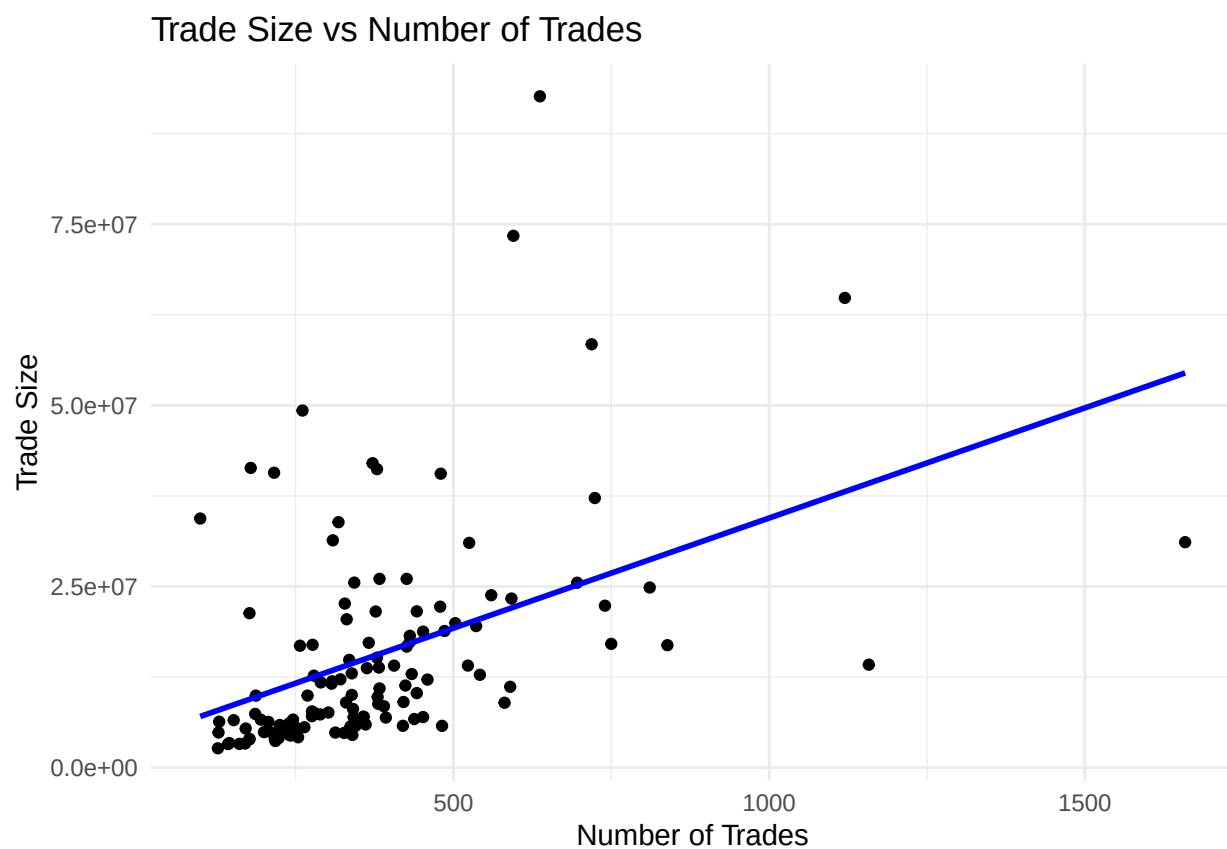
To assess whether a relationship exists between monthly trade volume and the number of transactions, both variables are plotted over time:





The plot raises some uncertainty about the relationship. Based on the analysis, the Spearman correlation between the number of trades per month and trade size per month is 0.6032014 and the Pearson correlation is 0.4444545, indicating a strong positive association.

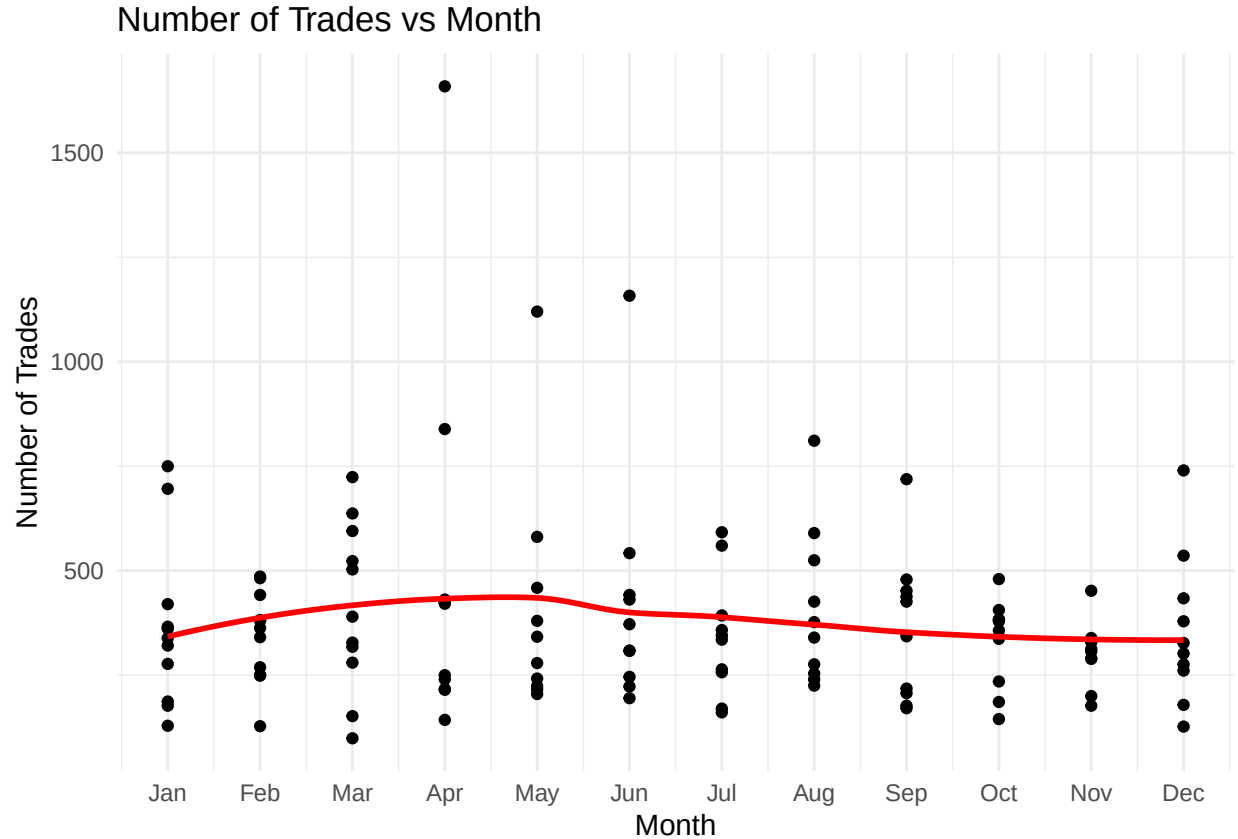
Overall, there is a strong correlation between monthly trade size and the number of trades. This is intuitive, as months with more transactions tend to exhibit higher total traded values. This relationship is further illustrated using a linear trend plot.



Next, we test whether the number of trades varies systematically across months or years.

The Spearman correlation per year is 0.0911164 and per month is -0.0642355, indicating no meaningful correlation.

The Spearman correlation between the number of trades and year, as well as between the number of trades and month, is close to zero. This indicates that trading activity does not exhibit significant seasonal or long-term time trends.



The corresponding plot confirms that the number of trades remains relatively stable across months. We can also compute the same for the trade size. The average trade size is 4.1135234×10^4

3) Chamber, party and excess return

3.1) Correlation and dominant party

We now analyze whether excess returns differ by political chamber or party.

The correlation between Chamber and excess return is -0.0128372, and between Chamber and net gain is -0.0151868.

For Party, the correlations are -0.0354317 and -0.0133452.

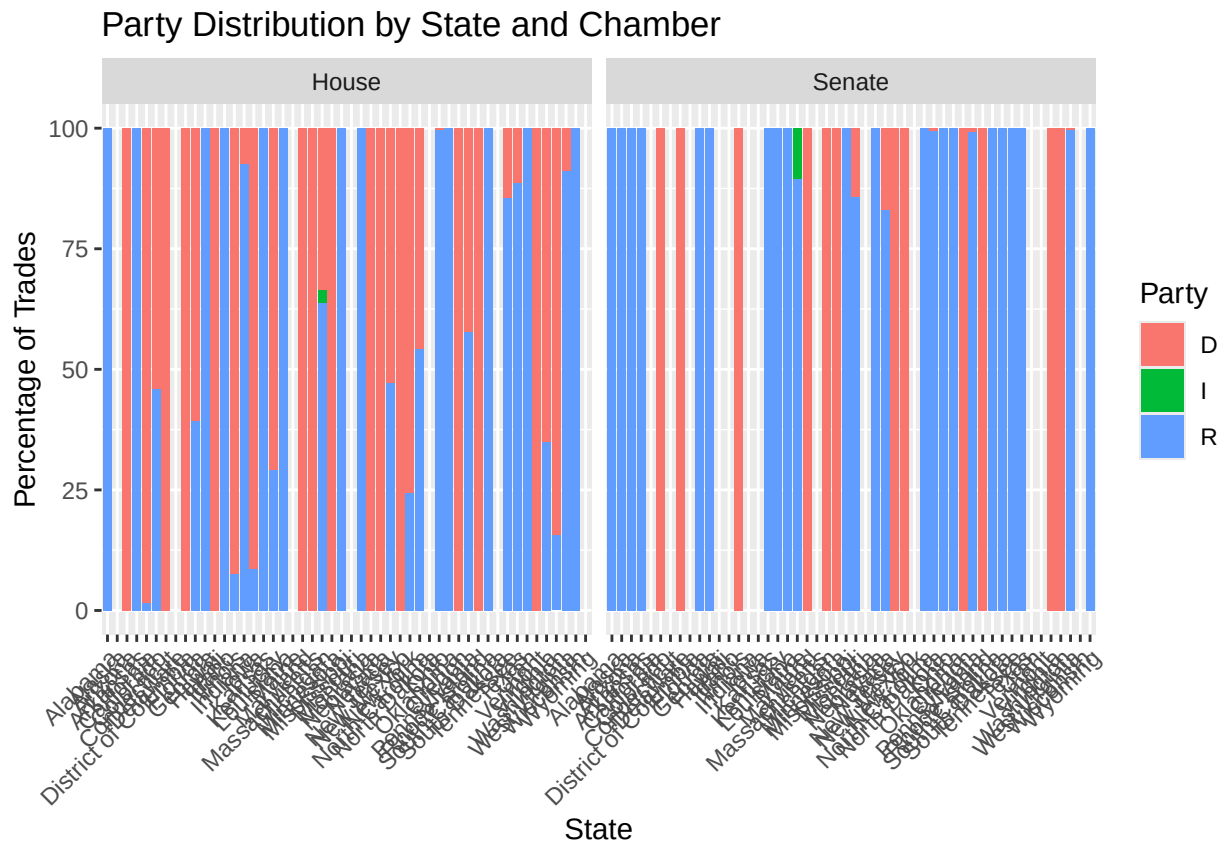
These results indicate very weak relationships between political chamber and excess return, as well as between party affiliation and excess return. Therefore, neither chamber nor party appears to be strongly associated with trading performance.

Next, trades are grouped by party and chamber within each state. The results are presented below.

```
## # A tibble: 6 x 4
##   State   Chamber Party Num_Trades
##   <chr>   <chr>   <chr>     <int>
## 1 Alabama House    R         371
## 2 Alabama Senate   R         976
## 3 Alaska  Senate   R         118
## 4 Arizona House    D         358
## 5 Arizona Senate   R          30
## 6 Arkansas House    R          45
```

Let us now examine the distribution of party affiliation across states.

```
## # A tibble: 6 x 4
##   State    Chamber Party Num_Trades
##   <chr>    <chr>   <chr>    <int>
## 1 Alabama House     R         371
## 2 Alabama Senate    R         976
## 3 Alaska  Senate    R         118
## 4 Arizona House     D        358
## 5 Arizona Senate    R          30
## 6 Arkansas House     R          45
```



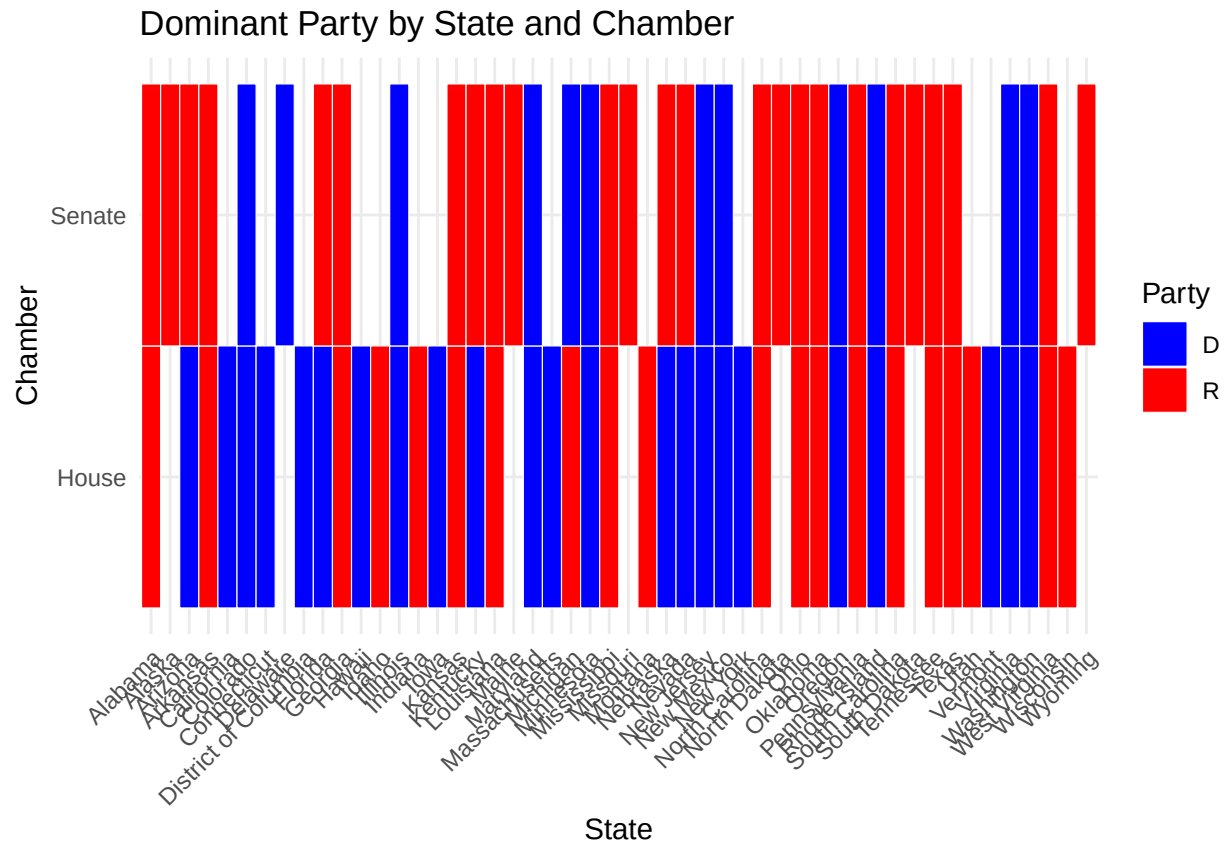
Next, we identify the dominant political party within each state and chamber using both heatmaps and geographic maps.

```
## # A tibble: 80 x 4
## # Groups:   State, Chamber [80]
##   State    Chamber Party Num_Trades
##   <chr>    <chr>   <chr>    <int>
## 1 Alabama House     R         371
## 2 Alabama Senate    R         976
## 3 Alaska  Senate    R         118
## 4 Arizona House     D        358
## 5 Arizona Senate    R          30
## 6 Arkansas House     R          45
## 7 Arkansas Senate    R          19
```

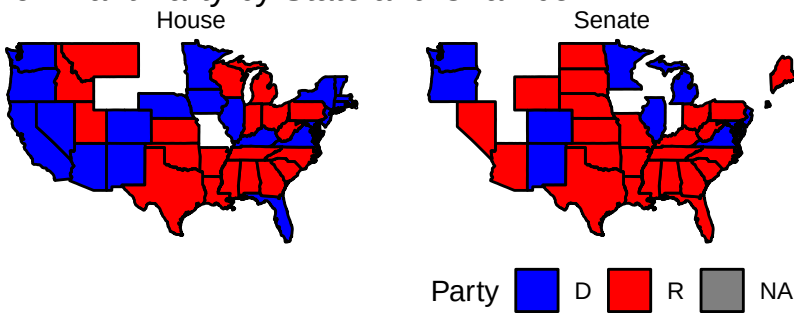
```
## 8 California House D 3588
## 9 Colorado House D 91
## 10 Colorado Senate D 88
## # i 70 more rows
```

```
## # A tibble: 80 x 4
## # Groups:   State, Chamber [80]
## State Chamber Party Num_Trades
## <chr> <chr> <chr> <int>
## 1 Alabama House R 371
## 2 Alabama Senate R 976
## 3 Alaska Senate R 118
## 4 Arizona House D 358
## 5 Arizona Senate R 30
## 6 Arkansas House R 45
## 7 Arkansas Senate R 19
## 8 California House D 3588
## 9 Colorado House D 91
## 10 Colorado Senate D 88
## 11 Connecticut House D 147
## 12 Delaware Senate D 751
## 13 District of Columbia House D 2
## 14 Florida House D 2358
## 15 Florida Senate R 6
## 16 Georgia House R 404
## 17 Georgia Senate R 2549
## 18 Hawaii House D 8
## 19 Idaho House R 140
## 20 Illinois House D 568
## 21 Illinois Senate D 22
## 22 Indiana House R 102
## 23 Iowa House D 203
## 24 Kansas House R 186
## 25 Kansas Senate R 641
## 26 Kentucky House D 366
## 27 Kentucky Senate R 28
## 28 Louisiana House R 19
## 29 Louisiana Senate R 165
## 30 Maine Senate R 349
## 31 Maryland House D 5
## 32 Maryland Senate D 10
## 33 Massachusetts House D 747
## 34 Michigan House R 559
## 35 Michigan Senate D 57
## 36 Minnesota House D 842
## 37 Minnesota Senate D 28
## 38 Mississippi House R 60
## 39 Mississippi Senate R 529
## 40 Missouri Senate R 6
## 41 Montana House R 1331
## 42 Nebraska House D 31
## 43 Nebraska Senate R 29
## 44 Nevada House D 1273
## 45 Nevada Senate R 69
```

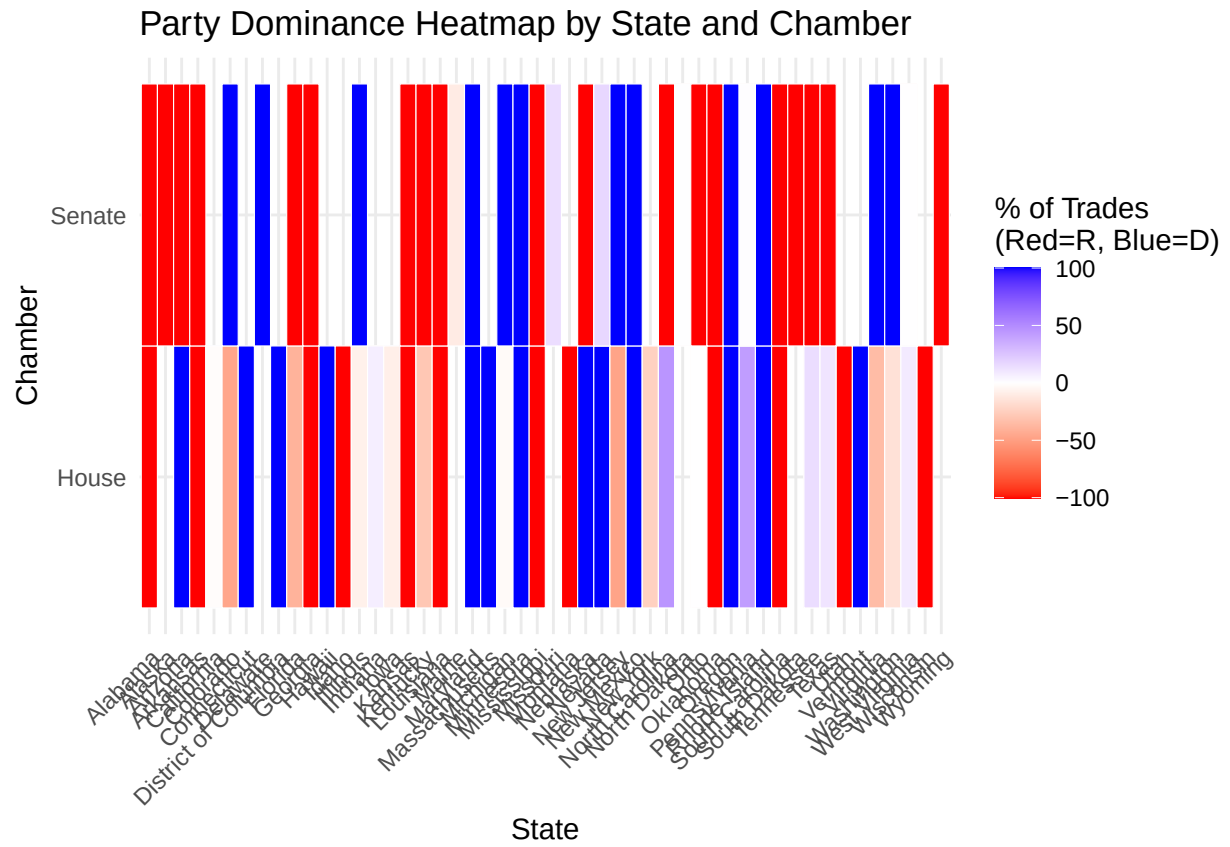

## 46 New Jersey	House	D	3209
## 47 New Jersey	Senate	D	13
## 48 New Mexico	House	D	1
## 49 New Mexico	Senate	D	2
## 50 New York	House	D	1928
## 51 North Carolina	House	R	930
## 52 North Carolina	Senate	R	98
## 53 North Dakota	Senate	R	184
## 54 Ohio	House	R	401
## 55 Ohio	Senate	R	1
## 56 Oklahoma	House	R	627
## 57 Oklahoma	Senate	R	365
## 58 Oregon	House	D	361
## 59 Oregon	Senate	D	240
## 60 Pennsylvania	House	R	213
## 61 Pennsylvania	Senate	R	150
## 62 Rhode Island	House	D	439
## 63 Rhode Island	Senate	D	887
## 64 South Carolina	House	R	76
## 65 South Carolina	Senate	R	5
## 66 South Dakota	Senate	R	1
## 67 Tennessee	House	R	621
## 68 Tennessee	Senate	R	958
## 69 Texas	House	R	3839
## 70 Texas	Senate	R	11
## 71 Utah	House	R	435
## 72 Vermont	House	D	102
## 73 Virginia	House	D	752
## 74 Virginia	Senate	D	11
## 75 Washington	House	D	315
## 76 Washington	Senate	D	167
## 77 West Virginia	House	R	564
## 78 West Virginia	Senate	R	442
## 79 Wisconsin	House	R	47
## 80 Wyoming	Senate	R	3



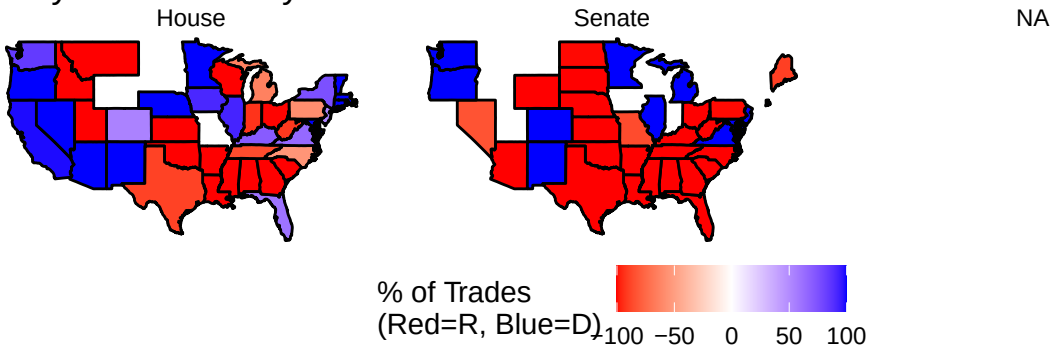
Dominant Party by State and Chamber



We now examine the intensity of party dominance across states.



Party Dominance by State and Chamber



3.2) Excess return by Party

```
## # A tibble: 3 x 5
##   Party Mean_Excess_Return Median_Excess_Return SD_Excess_Return Num_Trades
##   <chr>          <dbl>          <dbl>          <dbl>          <int>
## 1 D             -8.34            -22.6           207.          21882
## 2 I            -34.6            -52.2           102.           64
## 3 R            -21.2            -33.0           150.          24521
```

We now examine the statistical properties of excess returns.

```
summary(df$excess_return)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.    NA's
## -318.75  -84.40   -27.71   -15.03   16.10 10885.41  2506
```

The initial summary appears highly negative, which is primarily due to the presence of missing values. After excluding missing observations, the corrected statistics are as follows:

The mean excess return is: -15.0291822 The median excess return is: -27.7138947 The standard deviation is: 179.90644

Interpretation

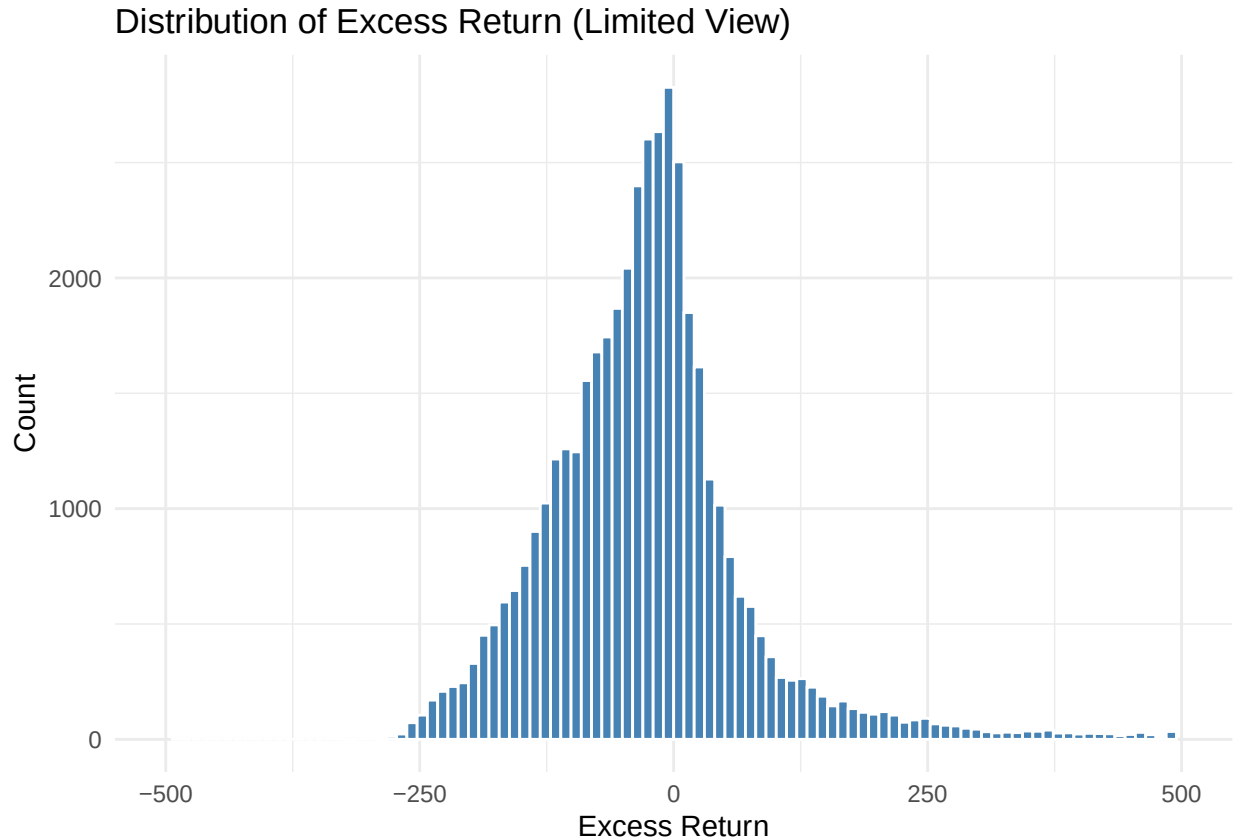
On average, trades underperformed the market benchmark by approximately 15%.

The mean is slightly higher than the median because a small number of large positive outcomes increase the average.

More than half of all trades underperformed the benchmark by roughly 28%, confirming a negatively skewed distribution.

The very high standard deviation indicates substantial volatility in trade outcomes, with extreme outliers such as 10,885 and −318 strongly influencing dispersion.

Distribution of excess returns - Histogram



The histogram suggests an approximately normal distribution with a pronounced right tail, indicating positive skewness driven by extreme high returns.

A negative value of `net_gain` represents a loss, while a positive value represents a gain. The top ten overperforming trades with highest returns based on net gain, are shown below.

##	Name	Party	Company	Ticker	net_gain
## 1	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	10885.406
## 2	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	10427.902
## 3	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	9432.152
## 4	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	9385.926
## 5	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	7852.900
## 6	Brenda Lulena Lawrence	D	NVIDIA CORPORATION	NVDA	3334.196
## 7	Debbie Dingell	D	NVIDIA CORPORATION	NVDA	3169.894
## 8	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	3115.648
## 9	Pete Sessions	R	NVIDIA CORPORATION	NVDA	3058.806
## 10	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	2422.490

#number of trades per ticker

```
ntpt <- df %>% group_by(Ticker) %>% summarise(Num_Trades = n())
ntpt
```

```
## # A tibble: 3,966 x 2
##   Ticker      Num_Trades
##   <chr>      <int>
## 1 07/01/28          1
## 2 07/01/36          1
## 3 OQZI.IL          2
## 4 12/01/37          1
## 5 13.WEEK MATURE    1
## 6 13.WEEK, MATURE    2
## 7 2                2
## 8 3%                1
## 9 3.MONTH, MATURE    2
## 10 3305779          2
## # i 3,956 more rows
```

```
max <-ntpt$Ticker[ntpt$Num_Trades == max(ntpt$Num_Trades)]
```

The most frequently traded ticker is MSFT.

Individual-level aggregation:

Trades were aggregated by individual politician to compute total activity and outcomes.

```
## # A tibble: 341 x 10
##   Name      Party State Chamber total_trades total_trade_value tot_excess_return
##   <chr>    <chr> <chr> <chr>      <int>          <dbl>          <dbl>
## 1 Thomas ~ R   New ~ House      2758      85266879      -118473.
## 2 Josh Go~ D   New ~ House      2574      264384287       21352.
## 3 Perdue,~ R   Geor~ Senate     2242      49400621      -59108.
## 4 Lois Fr~ D   Flor~ House     1358      11060679      -38700.
## 5 Greg Gi~ R   Mont~ House     1331      98100668.     -27173.
## 6 Susie L~ D   Neva~ House     1261      10440131       -9388.
## 7 Alan S.~ D   Cali~ House     1243      11299134.     -18951.
## 8 Daniel ~ D   New ~ House     1184      32031592.     -17808.
## 9 Gilbert~ D   Cali~ House     1142      12725071        1733.
## 10 K. Mich~ R   Texas House      977      16829488.     -21756.
## # i 331 more rows
## # i 3 more variables: Number_of_sale <int>, Number_of_purchase <int>,
## #   Net_gain <dbl>
```

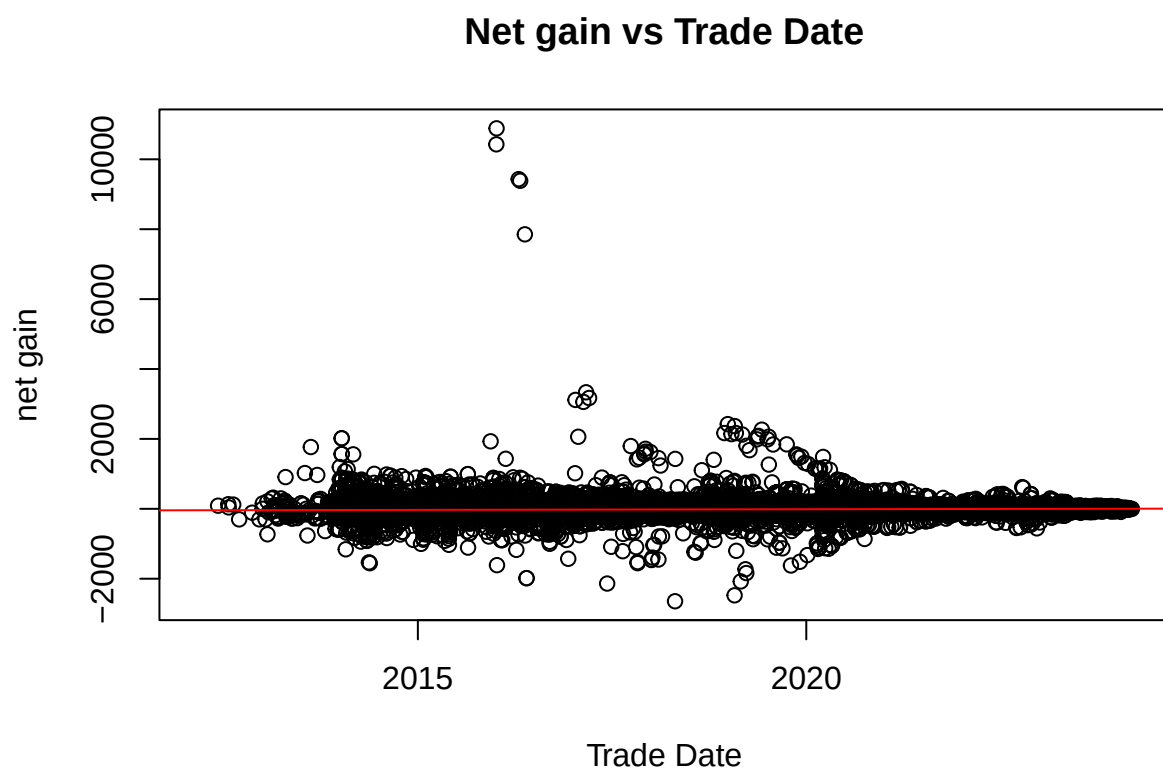
3.3) Time and performance

We test whether excess return is related to the timing of trades.

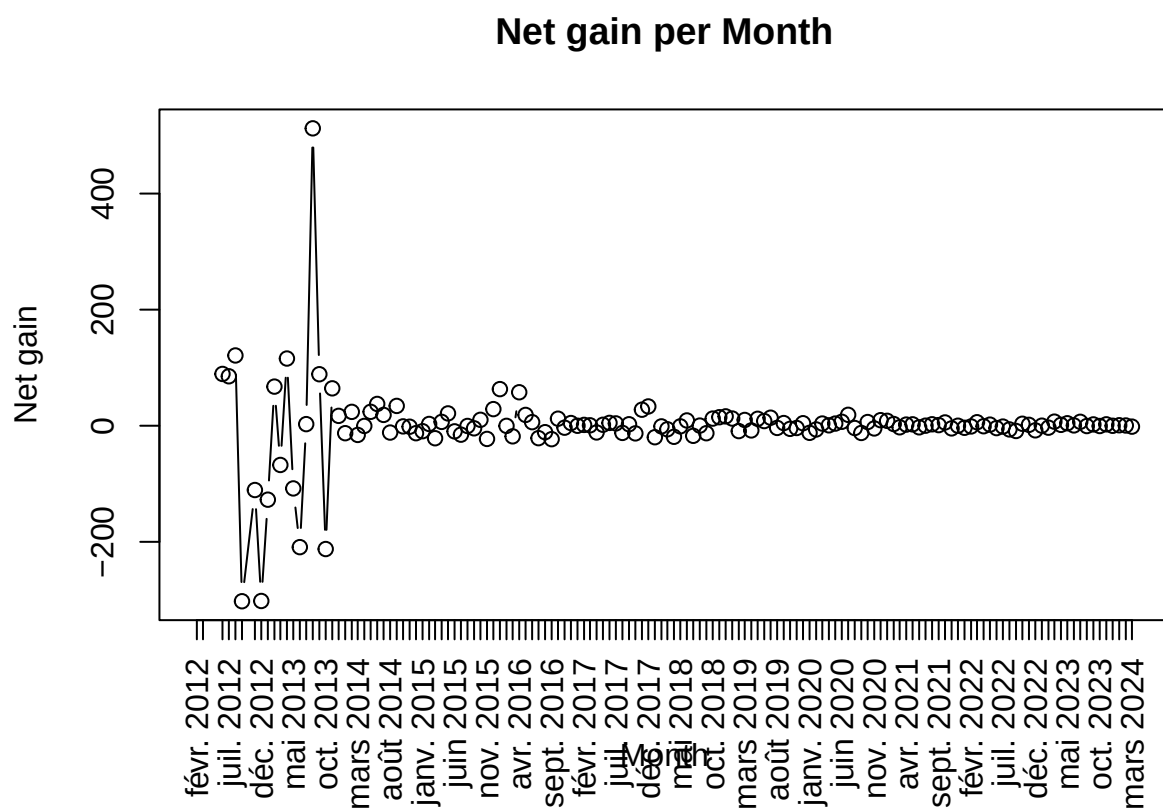
```
## [1] 0.05644884
```

The correlation is close to zero, indicating no direct relationship between trade date and excess return.

We further explore monthly and yearly patterns.



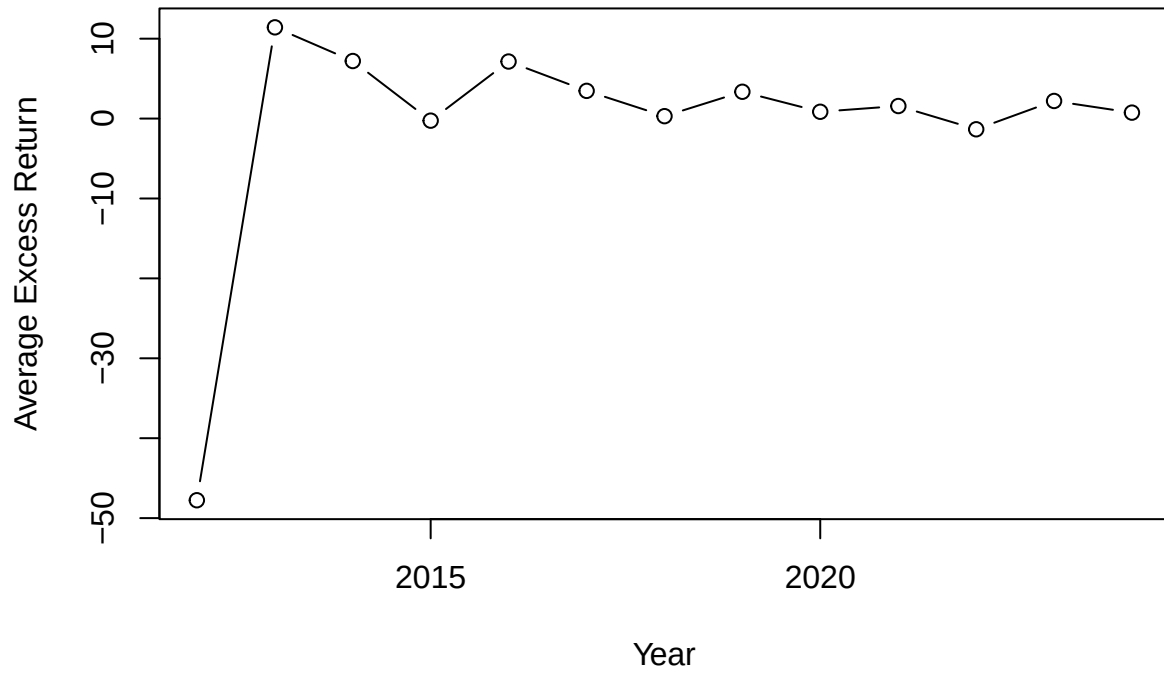
What is the relation between net gain and month?

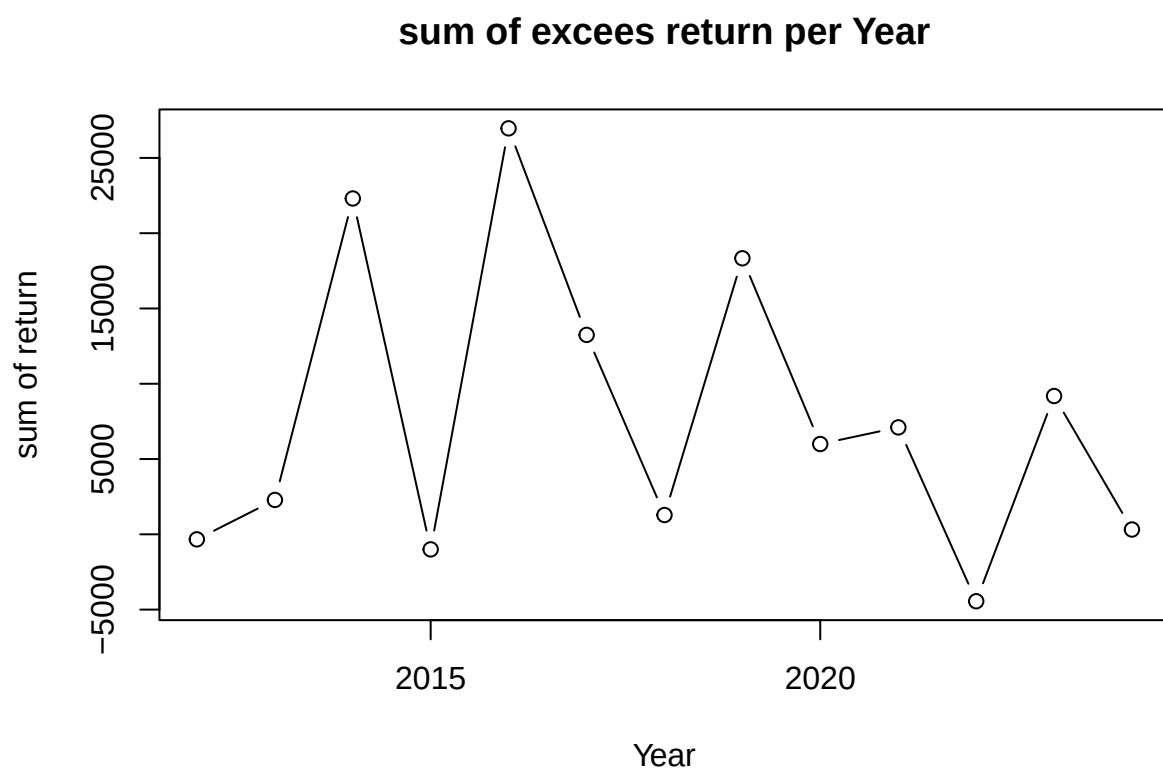


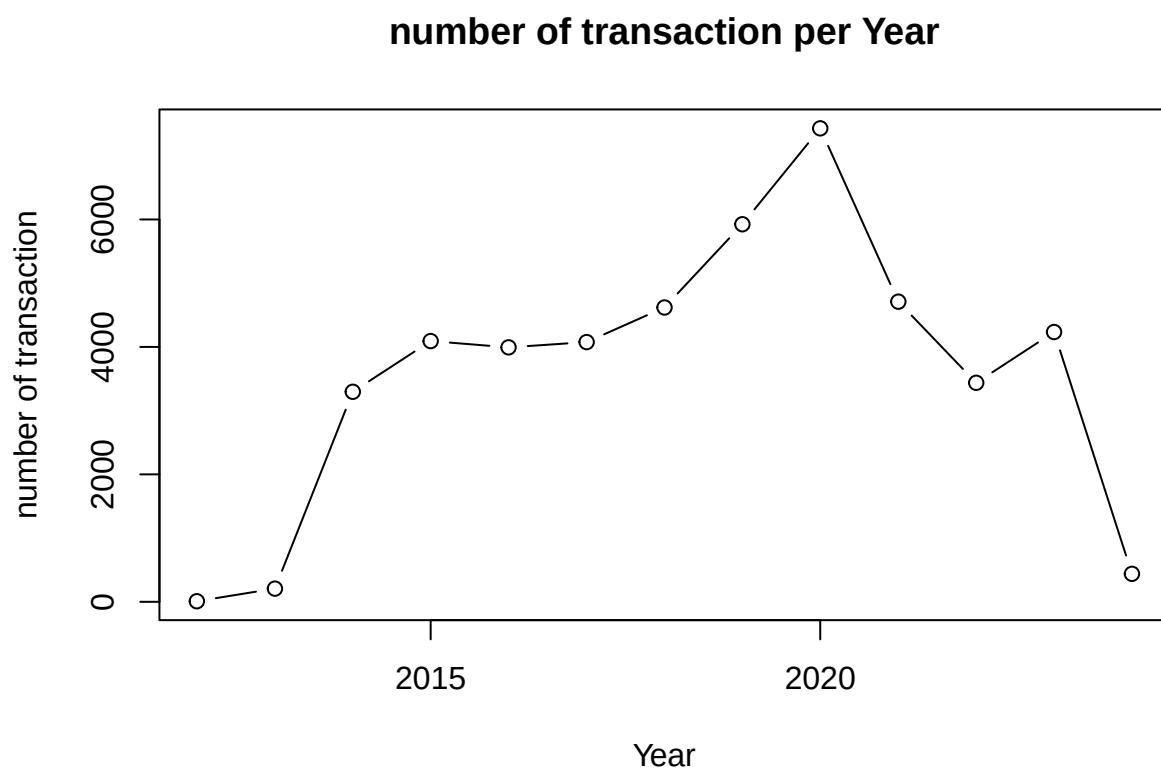
The average net gain shows a gradual increase over time but remains close to zero. Early years, particularly 2012, contain limited observations and should be interpreted with caution.

relation between excess of return and month

Average Excess Return per Year

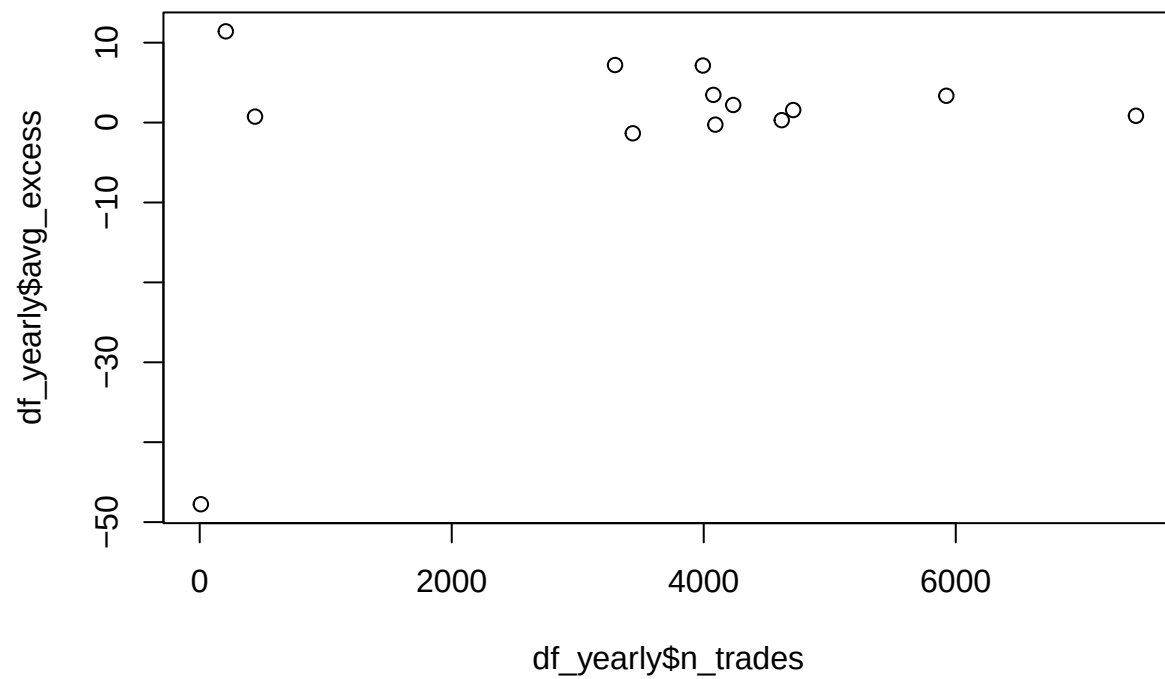






The average excess return increases over time. To avoid distortion from the low-activity year 2012, this year is excluded from further correlation analysis.

```
plot(df_yearly$n_trades, df_yearly$avg_excess)
```



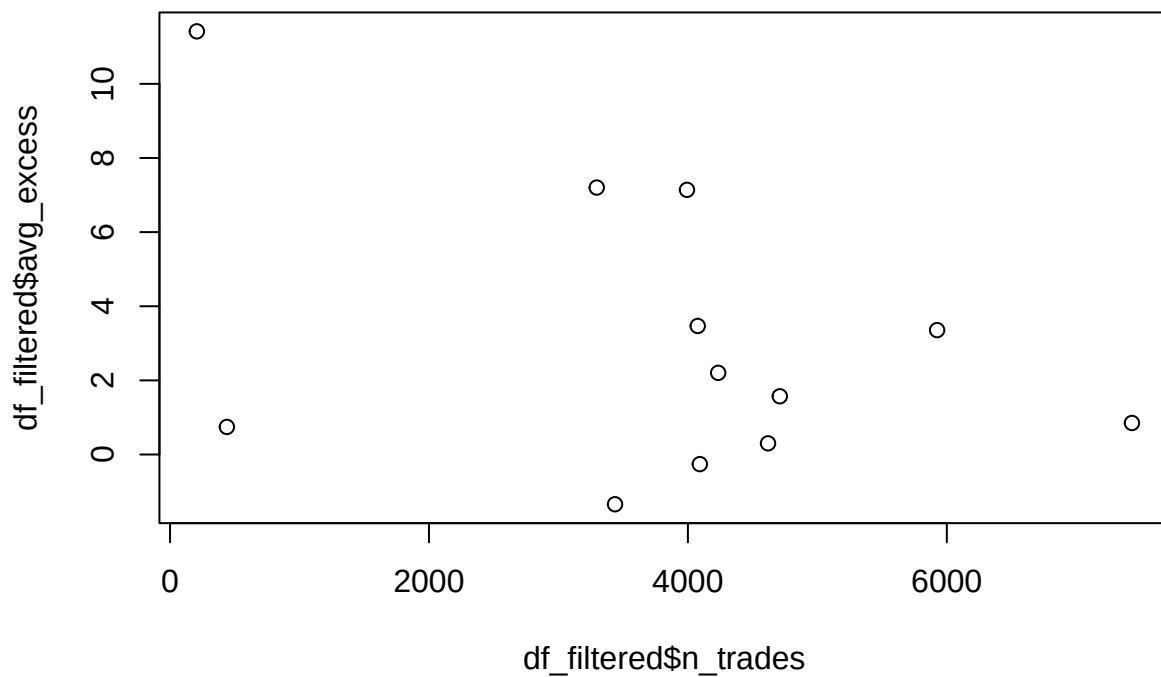
The point corresponding to 2012 biases the relationship.

The correlation is as follows:

```
## [1] -0.4131439
```

A modest positive relationship is observed between the number of trades and average net gain.

```
plot(df_filtered$n_trades, df_filtered$avg_excess)
```



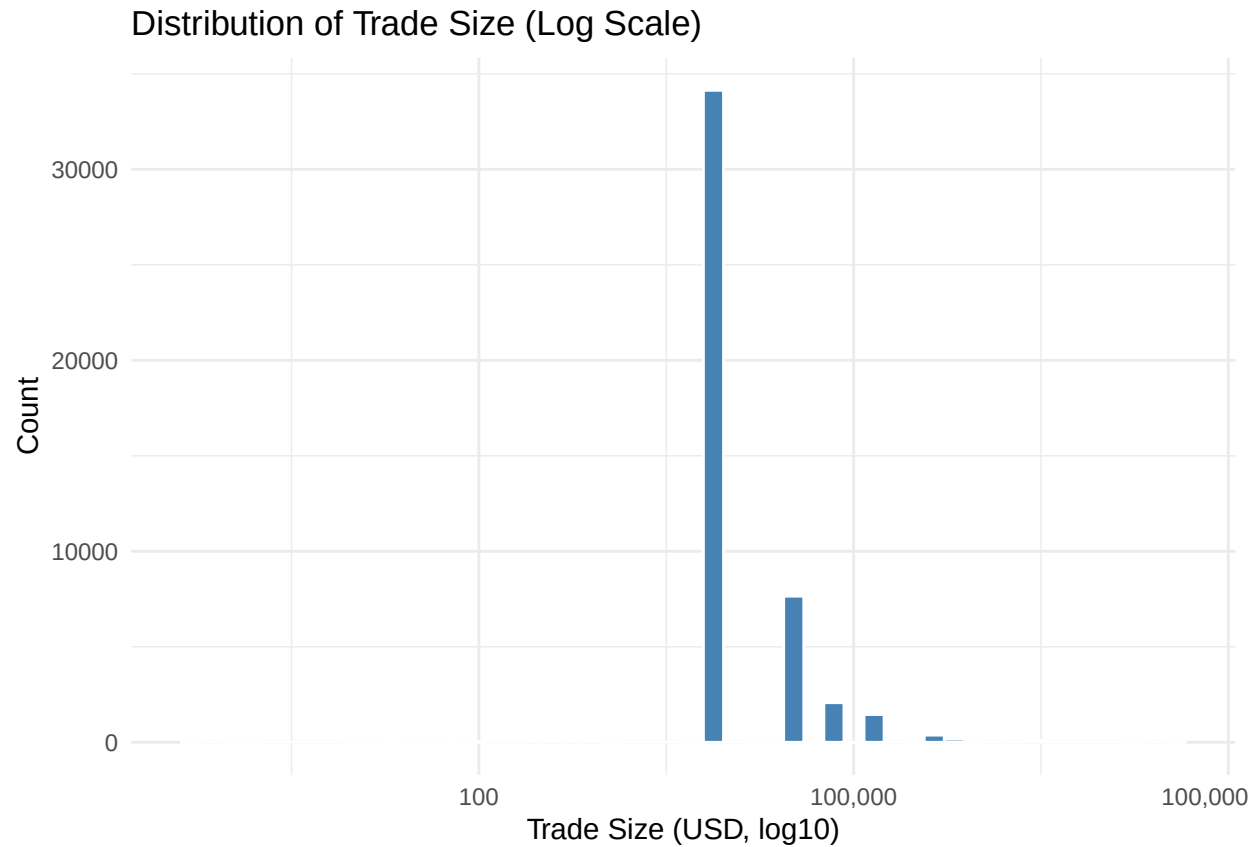
3.4) excess return, trade size by party

##1. Histogram of trade size

We begin by examining the distribution of trade size. Because trade sizes are highly skewed, we visualize the distribution on a logarithmic scale.

#LOG scale

```
ggplot(df, aes(x = trade_size_mid)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  scale_x_log10(labels = scales::comma) +
  labs(
    title = "Distribution of Trade Size (Log Scale)",
    x = "Trade Size (USD, log10)",
    y = "Count"
  ) +
  theme_minimal()
```

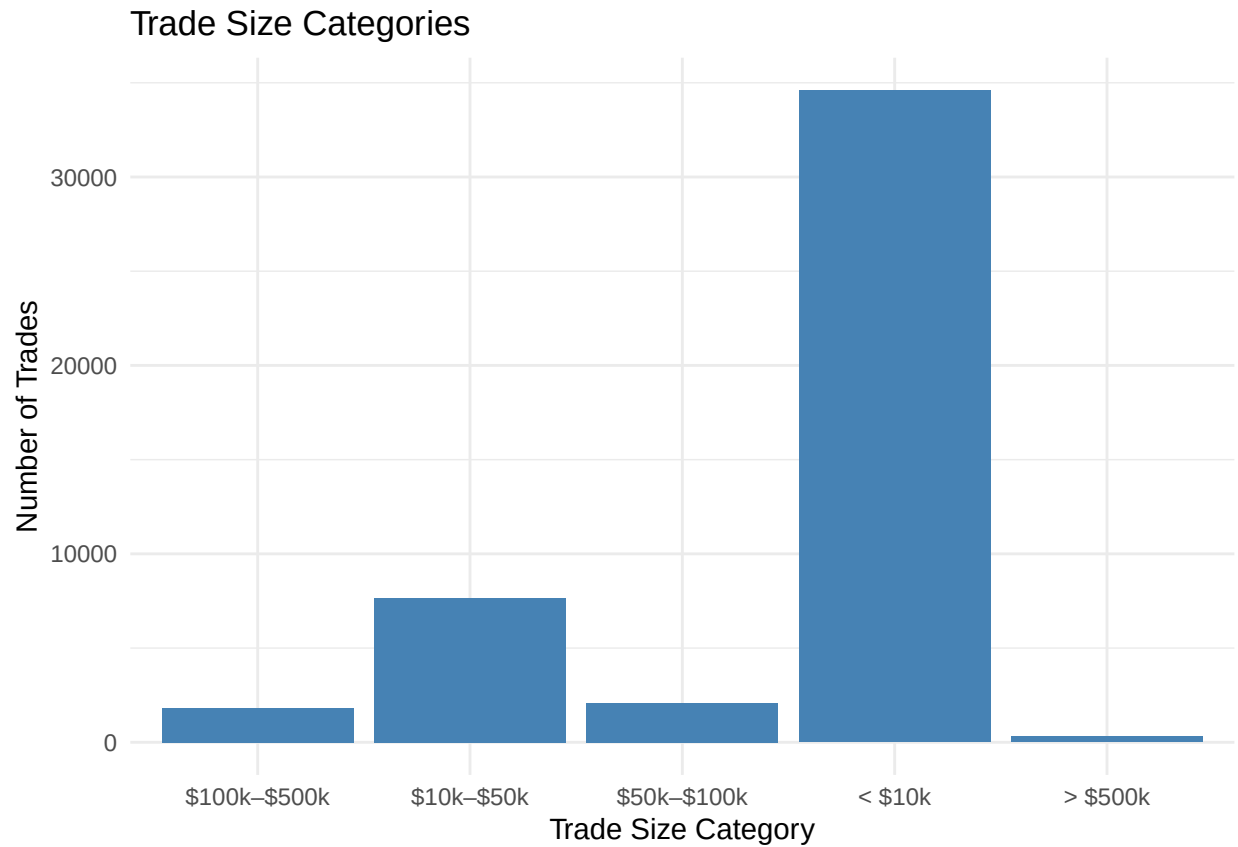


To improve interpretability, we also group trades into size categories.

```
#categorising trade size

df <- df %>%
  mutate(trade_size_cat = case_when(
    trade_size_mid < 1e4 ~ "< $10k",
    trade_size_mid < 5e4 ~ "$10k-$50k",
    trade_size_mid < 1e5 ~ "$50k-$100k",
    trade_size_mid < 5e5 ~ "$100k-$500k",
    TRUE ~ "> $500k"
  ))

ggplot(df, aes(x = trade_size_cat)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Trade Size Categories",
    x = "Trade Size Category",
    y = "Number of Trades"
  ) +
  theme_minimal()
```

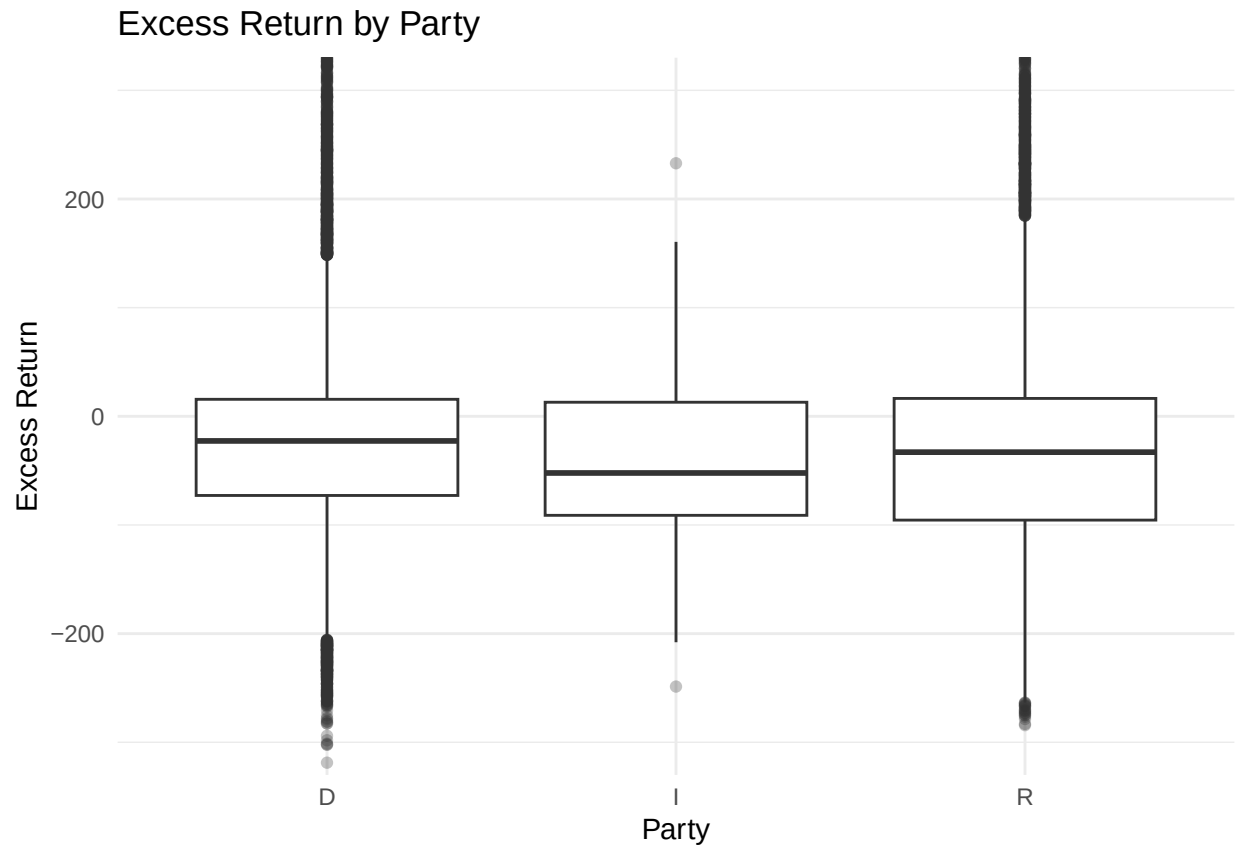


##2. Boxplot: excess return by party

We next compare the distribution of excess returns across political parties.

```
ggplot(df, aes(x = Party, y = excess_return)) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_cartesian(ylim = c(-300, 300)) +
  labs(
    title = "Excess Return by Party",
    x = "Party",
    y = "Excess Return"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2506 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

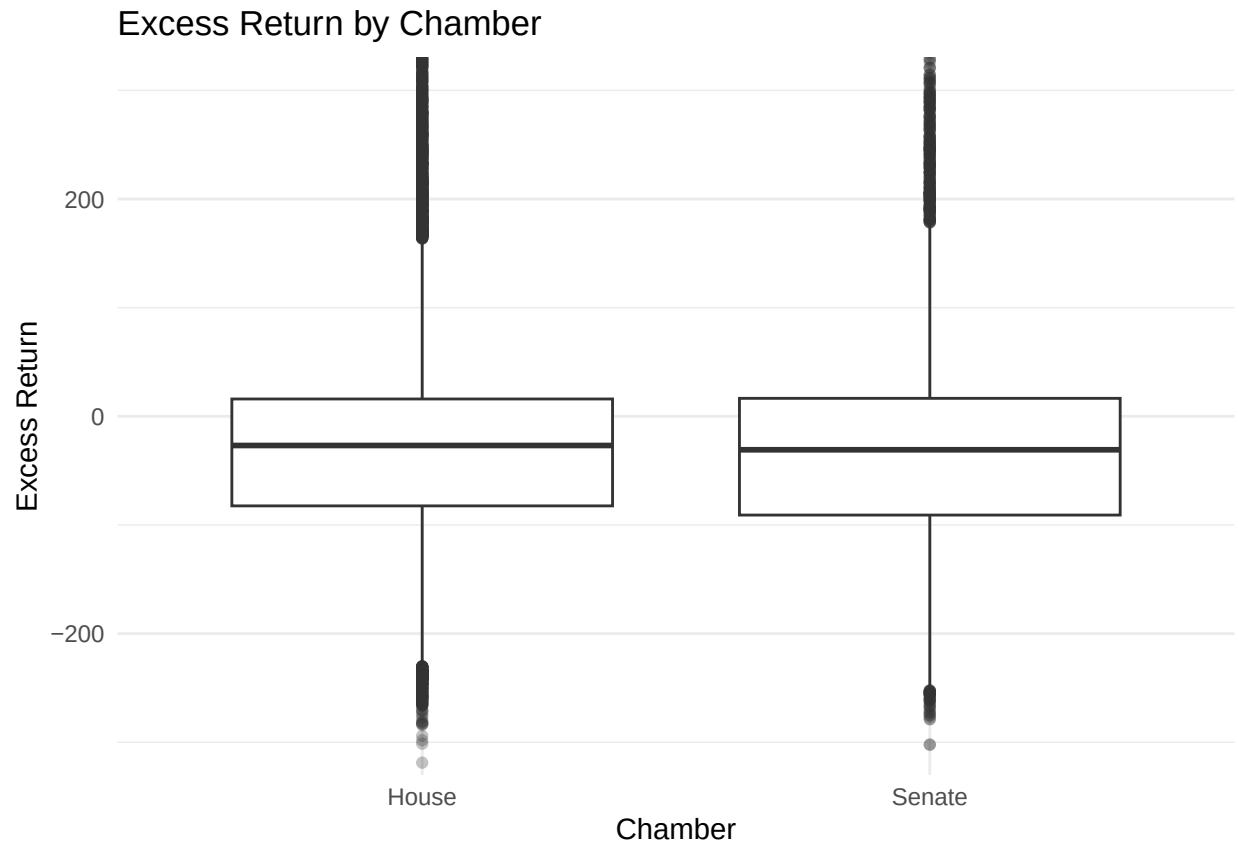



##3.1 Boxplot: excess return by chamber

We also examine whether excess returns differ between the House and the Senate.

```
ggplot(df, aes(x = Chamber, y = excess_return)) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_cartesian(ylim = c(-300, 300)) +
  labs(
    title = "Excess Return by Chamber",
    x = "Chamber",
    y = "Excess Return"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2506 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



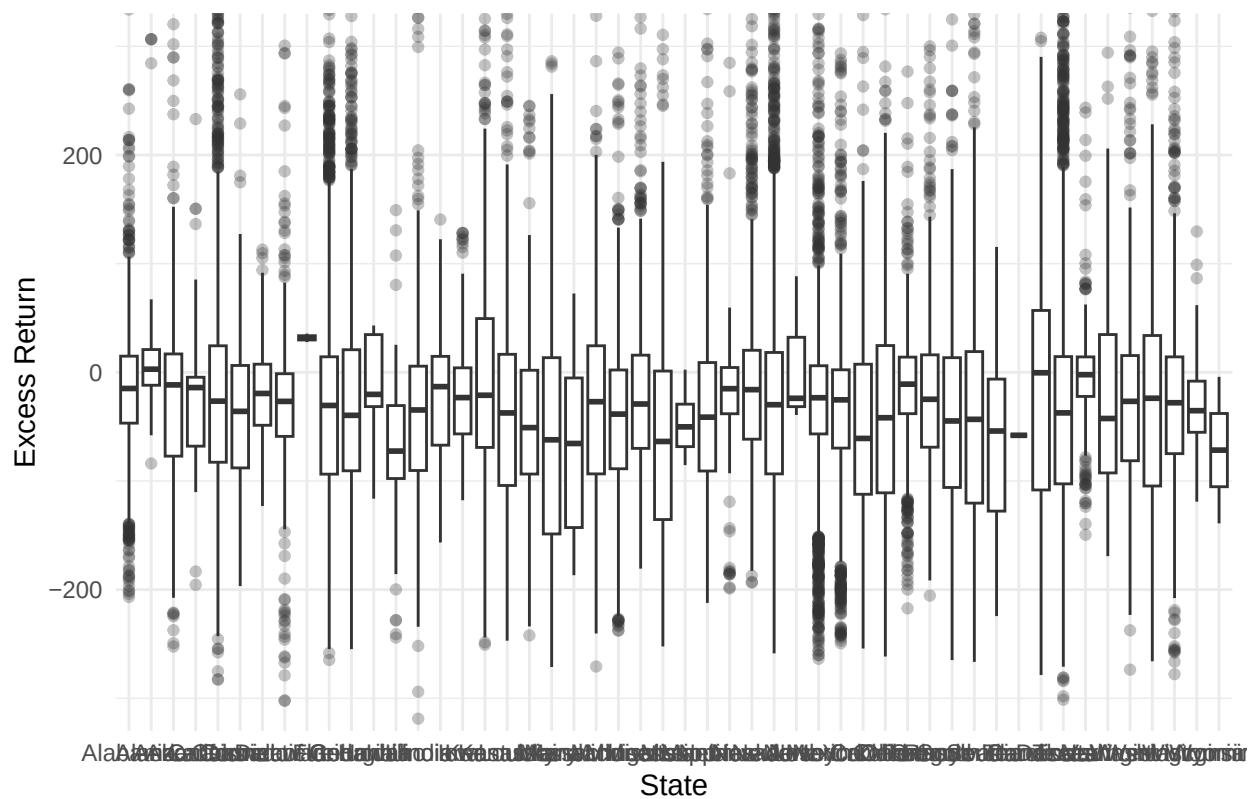
##3.2 excess return by state/ net gain by state

To explore geographic variation, we compare excess return distributions across states and compute state-level averages for both excess return and net gain.

```
ggplot(df, aes(x = State, y = excess_return)) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_cartesian(ylim = c(-300, 300)) +
  labs(
    title = "Excess Return by State",
    x = "State",
    y = "Excess Return"
  ) +
  theme_minimal()
```

```
## Warning: Removed 2506 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Excess Return by State



```
state_means <- df %>%
  group_by(State) %>%
  summarise(mean_excess_return = mean(excess_return, na.rm = TRUE))

state_means2 <- df %>%
  group_by(State) %>%
  summarise(net_gain = mean(net_gain, na.rm = TRUE))

us_states <- map_data("state")

state_means <- state_means %>%
  mutate(State = tolower(State))

state_means2 <- state_means2 %>%
  mutate(State = tolower(State))

map_df <- left_join(
  us_states,
  state_means,
  by = c("region" = "State")
)

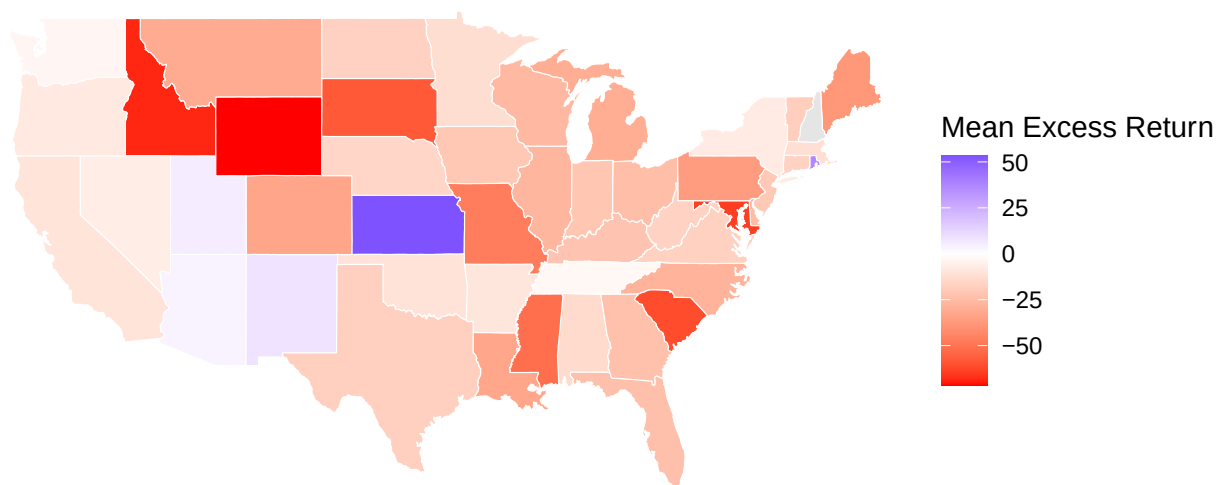
ggplot(map_df, aes(x = long, y = lat, group = group, fill = mean_excess_return)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
```

```

scale_fill_gradient2(
  low = "red",
  mid = "white",
  high = "blue",
  midpoint = 0,
  na.value = "grey90"
) +
labs(
  title = "Mean Excess Return by State",
  fill = "Mean Excess Return"
) +
theme_void()

```

Mean Excess Return by State



```

map_df2 <- left_join(
  us_states,
  state_means2,
  by = c("region" = "State")
)

ggplot(map_df2, aes(x = long, y = lat, group = group, fill = net_gain)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",

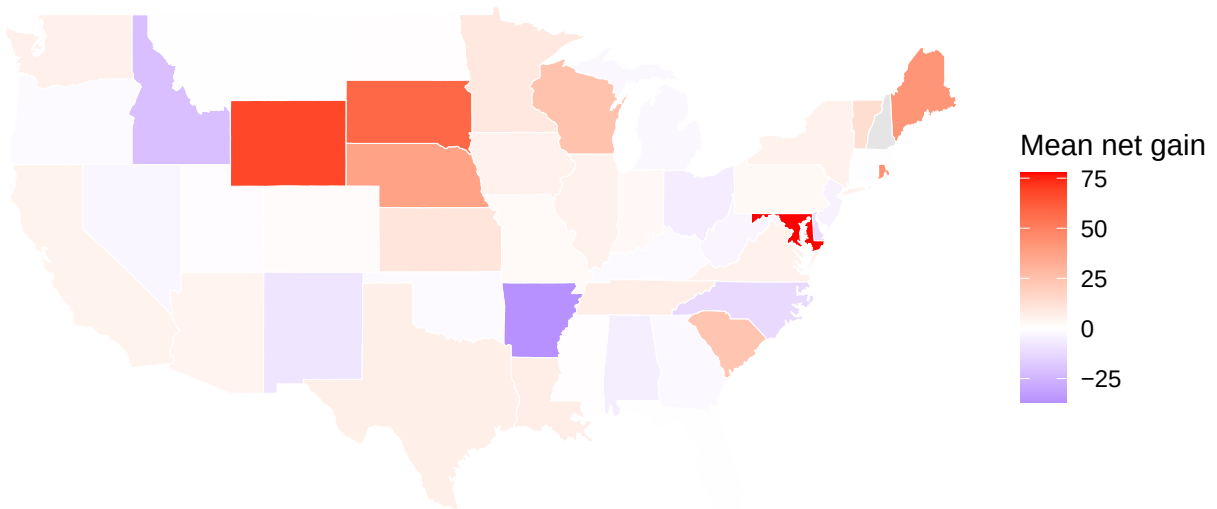
```

```

    high = "red",
    midpoint = 0,
    na.value = "grey90"
  ) +
  labs(
    title = "net gain by state",
    fill = "Mean net gain"
  ) +
  theme_void()

```

net gain by state



##3.3 Boxplot: excess return by party

To visualize the dispersion and individual observations simultaneously, we overlay points on the party-level boxplots.

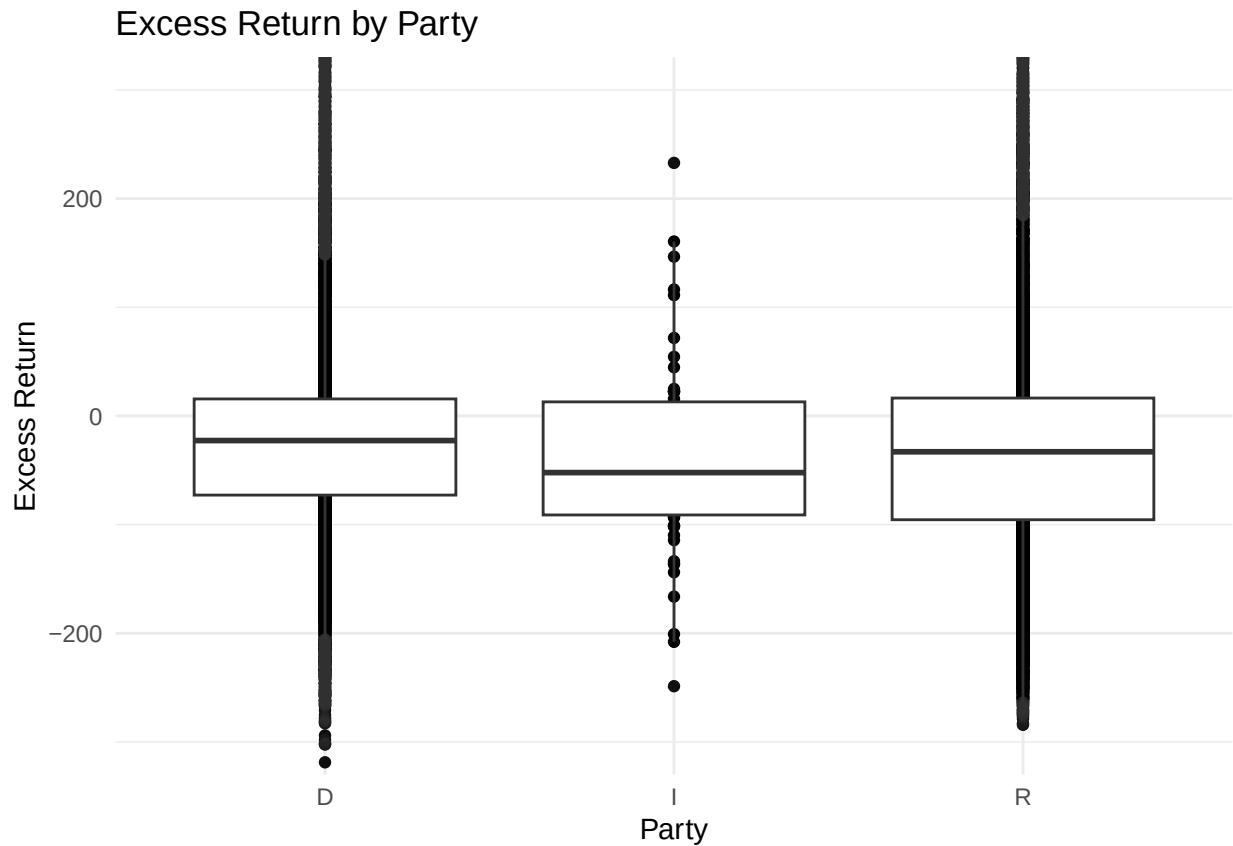
```

ggplot(df, aes(x = Party, y = excess_return)) + geom_point(na.rm = TRUE) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_cartesian(ylim = c(-300, 300)) +
  labs(
    title = "Excess Return by Party",
    x = "Party",
    y = "Excess Return"
  ) +
  theme_minimal()

```

Warning: Removed 2506 rows containing non-finite outside the scale range

```
## (`stat_boxplot()`).
```



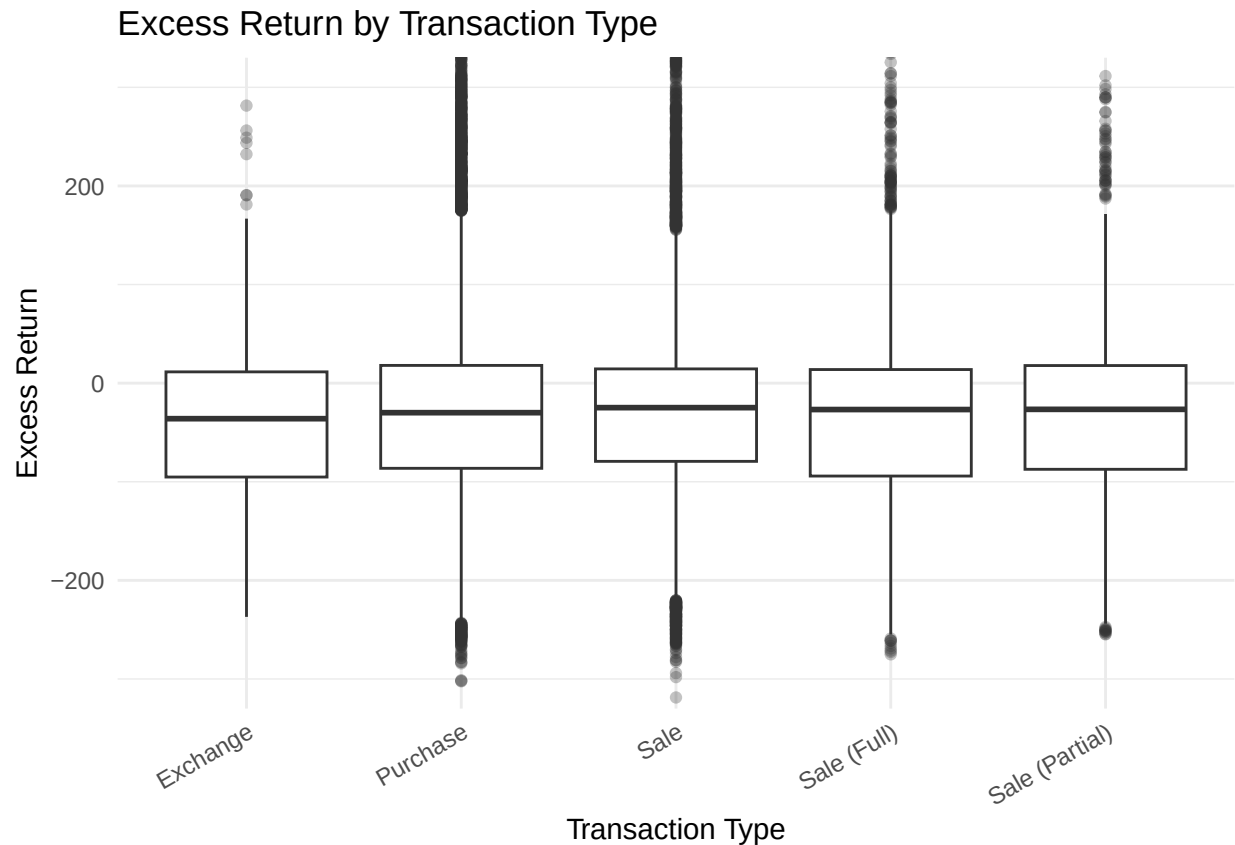
##4. Boxplot: excess return by transaction type

We further compare excess returns across transaction types (e.g., purchases and sales).

```
ggplot(df, aes(x = Transaction, y = excess_return)) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_cartesian(ylim = c(-300, 300)) +
  labs(
    title = "Excess Return by Transaction Type",
    x = "Transaction Type",
    y = "Excess Return"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```

```
## Warning: Removed 2506 rows containing non-finite outside the scale range
```

```
## (`stat_boxplot()`).
```

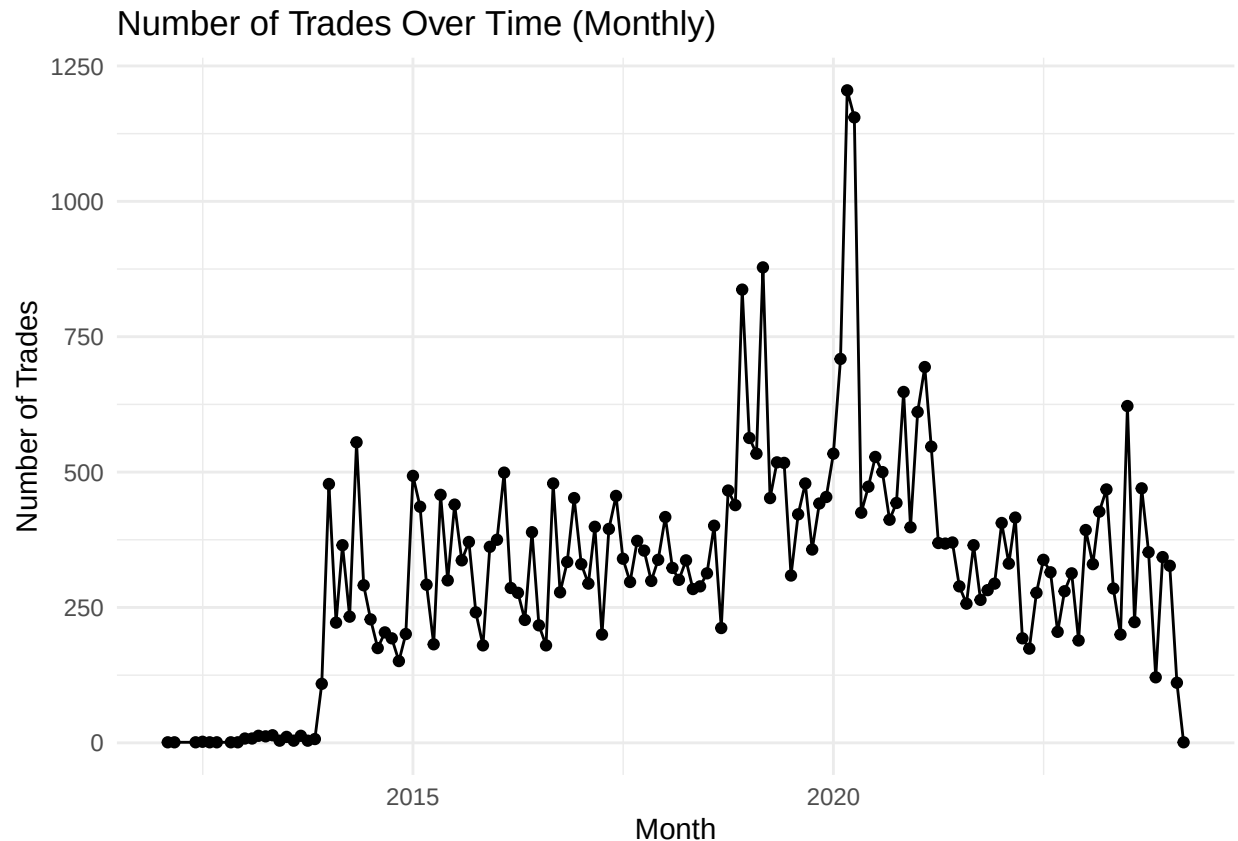


##6. Clean time series: number of trades over time

Finally, we construct a monthly time series of the number of trades to illustrate how trading activity evolves over time.

```
df_ts <- df %>%
  mutate(Month = as.Date(paste0(format(Traded, "%Y-%m"), "-01"))) %>%
  group_by(Month) %>%
  summarise(num_trades = n(), .groups = "drop")

ggplot(df_ts, aes(x = Month, y = num_trades)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Number of Trades Over Time (Monthly)",
    x = "Month",
    y = "Number of Trades"
  ) +
  theme_minimal()
```



Correlation between trade size and party

We test whether total traded value differs between parties using a two-sample t-test.

```
df_two <- df_person |> filter(Party %in% c("R", "D"))
```

```
t.test(total_trade_value ~ Party, data = df_two)
```

```
##
##  Welch Two Sample t-test
##
## data:  total_trade_value by Party
## t = 0.22042, df = 242.48, p-value = 0.8257
## alternative hypothesis: true difference in means between group D and group R is not equal to 0
## 95 percent confidence interval:
##  -4255936  5328439
## sample estimates:
## mean in group D mean in group R
##      5938708      5402457
```

The test suggests no statistically significant difference between the two parties; trade values appear broadly balanced.

3.5) Sales and purchases

We summarise total traded value by transaction type and report the number of sales, purchases, and exchanges.


```

df_sale <- df %>%
  group_by(Transaction)%>%
  summarise(Trade_size = sum(trade_size_mid))
View(df_sale)

sum(data$Transaction == "Sale" | data$Transaction == "Sale (Full)" | data$Transaction == "Sale (Partial)

## [1] 23045

sum(data$Transaction == "Purchase", na.rm = TRUE)

## [1] 23013

sum(data$Transaction == "Exchange", na.rm = TRUE)

## [1] 409

```

A correlation comparison between sales and purchases is not meaningful because the number of observations differs substantially between these transaction types.